Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/04/2006
PI Name:	Ahlers, Guenter Ph.D.		
Project Title:	Boundary Effects on Transport Properties and Dynamic Finite-Size Scaling Near the Superfluid-Transition Line of 4He		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	805-893-3795
Organization Name:	University of California		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:	http://www.nls.physics.ucsb.edu/		
City:	Santa Barbara	**State:**	CA
Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007, per N. Hall-GRC (9/06).		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Liu, Feng-Chuan (JPL)		
Grant/Contract No.:	NAG 3 2872		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	The task objective will be accomplished by measuring the thermal conductivity of liquid helium confined in cylindrical tubes of diameters from 1 to 100 microns. The objective of BEST is to determine the effects of boundaries on the thermal conductivity of liquid helium-4 near the superfluid transition The result of this work will provide the first test of scaling for transport properties near a continuous phase transition.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	A comparison between experimental results and theoretical prdictions will enhance our ability to calculate properties of complex systems using renormalization-group and Monte Carlo methods.
Task Progress:	Thermal-conductivity measurements were made on plates with channels of 1x10 micro-m cross section, and simultaneously on bulk helium. The bulk measurements served to provide the bulk lambda-point as a fixed point on the temperature scale. Analysis of the data is now complete and a publication has been submitted to the Journal of Low Temperature Physics. To our great surprise the results show that the finite-size scaling-function for this geometry does not differ measurably from the scaling function for cylindrical confinement. This differs qualitatively from what had been found for thermodynamic properties in one-dimensional and two-dimensional confinement.
Bibliography Type:	Description: (Last Updated: 02/06/2013)
Articles in Peer-reviewed Journals	Jerebets S, Liu Y, Liu F, Ahlers G. "Finite-size effects on the thermal resistivity of 4He near the superfluid transition." Journal of Low Temperature Physics. 2007 Mar;146(5-6):471-83. http://dx.doi.org/10.1007/s10909-006-9278-5 , Mar-2007